

4 Recurrence

Substitution Example

$$\begin{aligned}T(n) &= cn \\T(n+1) &= c(n+1) \\&= T(n) + c \\&= cn + c \\&= c(n+1)\end{aligned}$$

Recursion Tree

$$2T(n/2) + n^2.$$

Master Theorem

Check given $T(n)$ is the form of $aT(n/b) + f(n)$

$$T(n) = 8T(n/2) + n^2 \tag{1}$$

$a = 8, b = 2, f(n) = n^2$. Prove $f(n) \in O(n^{\log_2 a - \epsilon}) = O(n^{3-\epsilon}) = O(n^{3-1}) = O(n^2)$. $T(n) = \Theta(n^{\log_2 a}) = \Theta(n^3)$